

Section 4.3 Derivatives and Shapes of Graphs

53) Use derivatives to determine where the function is increasing, decreasing, concave up, concave down and find all minimum and maximum values.

$$f(x) = \ln(x^4 + 8)$$

$$f'(x) = \frac{4x^3}{x^4 + 8}$$

$$f'(x) = 0$$

$$4x^3 = 0$$

$$x^3 = 0$$

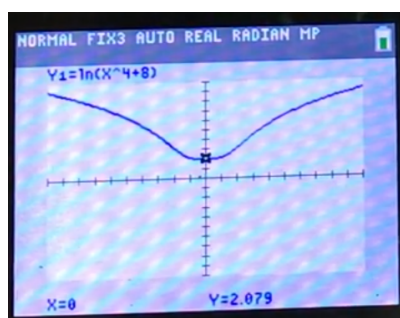
$$x = 0$$

$$f'' \text{ DNE}$$

$$x^4 + 8 = 0$$

$$x^4 = -8$$

$$x = \sqrt[4]{-8} \text{ imaginary}$$



$$\text{Min } (0, 2.079)$$

$$\text{Decr. } (-\infty, 0)$$

$$\text{Incr. } (0, \infty)$$

$$f''(x) = \frac{12x^2(x^4 + 8) - 4x^3(4x^3)}{(x^4 + 8)^2} = 0$$

$$0 = 4x^2 [3x^4 + 24 - 4x^4]$$

$$0 = 4x^2 [-x^4 + 24]$$

$$4x^2 = 0$$

$$x^2 = 0$$

$$x = 0$$

$$-x^4 + 24 = 0$$

$$24 = x^4$$

$$\pm \sqrt[4]{24} = \sqrt[4]{x}$$

$$\pm 2.213 = x$$

inflection point

Concave down

$$(-\infty, -2.213) \cup (2.213, \infty)$$

Concave up

$$(-2.213, 2.213)$$

54) Use derivatives to determine where the function is increasing, decreasing, concave up, concave down and find all minimum and maximum values.

$y = \sin x - \cos x$ on the interval $[0, 2\pi)$

$$y' = \cos x + \sin x$$

$$0 = \cos x + \sin x$$

$$\frac{-\sin x}{\cos x} = \frac{\cos x}{\cos x}$$

$$-\tan x = 1$$

$$\tan x = -1$$

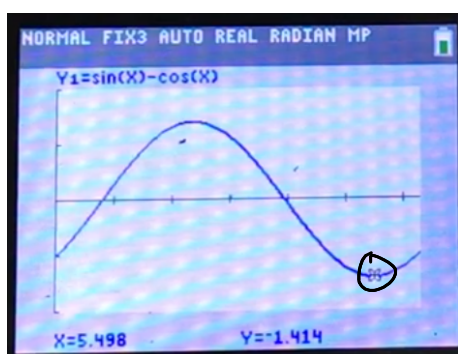
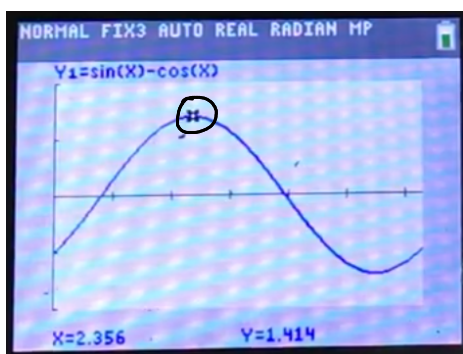
$$x = \tan^{-1}(-1)$$

$$x = \frac{3\pi}{4}$$

$$x = \frac{7\pi}{4}$$

$$\text{Max } (2.356, 1.414)$$

$$\text{Min } (5.498, -1.414)$$

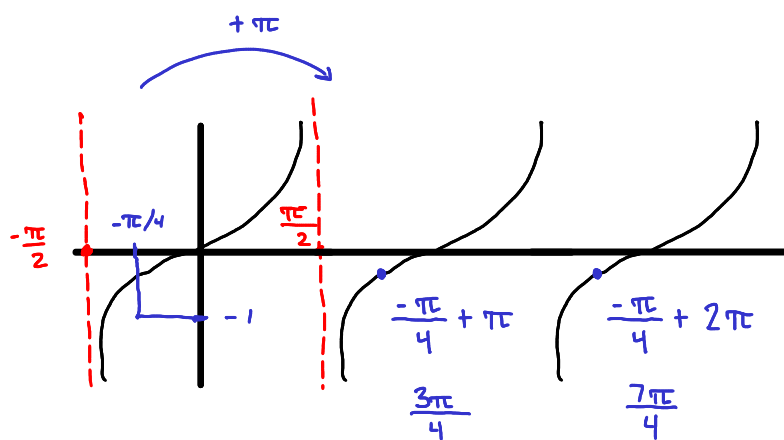


$$\text{Incr. } \left[0, \frac{3\pi}{4}\right) \cup \left(\frac{7\pi}{4}, 2\pi\right]$$

$$\text{Decr. } \left(\frac{3\pi}{4}, \frac{7\pi}{4}\right)$$

$$\text{Concave up } \left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right)$$

$$\text{Concave down } \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$$



$$y' = \cos x + \sin x$$

$$y'' = -\sin x + \cos x$$

$$0 = -\sin x + \cos x$$

$$\frac{\sin x}{\cos x} = \frac{\cos x}{\cos x}$$

$$\tan x = 1$$

$$x = \frac{\pi}{4} \checkmark$$

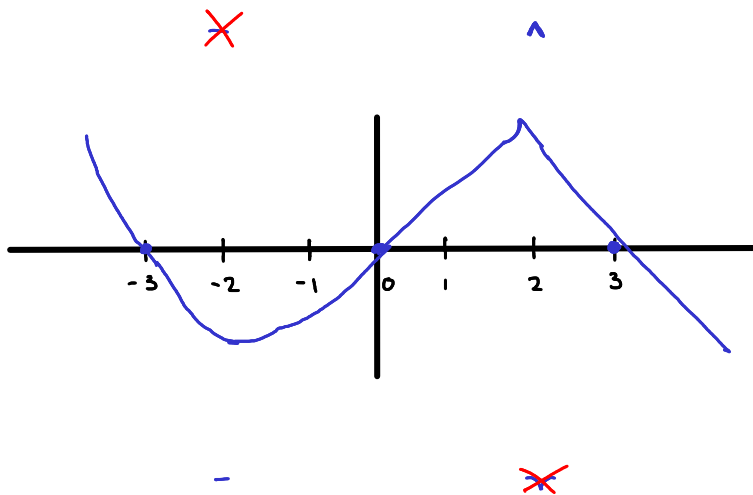
$$x = \frac{5\pi}{4} \checkmark$$

55) Sketch the graph of the continuous function $f(x)$ with all of the following properties:

$$f(-3) = f(0) = f(3) = 0$$

$$f'(-2) = 0, \underline{f'(2) \text{ DNE}}, f'(x) > 0 \text{ on the interval } (-2, 2)$$

$$f''(x) > 0 \text{ on } (-3, 2), \underline{f''(x) = 0 \text{ on } [2, \infty)}$$



Side Note

If $f'(2)$ DNE then $f''(2)$ cannot exist
but saying $f''(x) = 0$ on $[2, \infty)$ includes $x = 2$

$$f''(x) = 0 \text{ for } x > 2 \quad \checkmark$$

Section 4.4 L'Hospital's Rule

56) Evaluate $\lim_{x \rightarrow 0} \frac{x - \sin x}{x - \tan x}$.

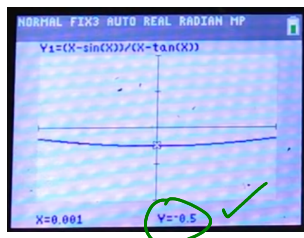
$\frac{0}{0}$

$$1'H = \lim_{x \rightarrow 0} \frac{1 - \cos x}{1 - \sec^2 x}$$

$$1'H = \lim_{x \rightarrow 0} \frac{\sin x}{-2 \sec x \cdot \sec x \cdot \tan x}$$

$$= \frac{\cancel{\sin x} \cdot \frac{1}{\cancel{\sin x}}}{-2 \cdot \frac{1}{\cos^2 x} \cdot \frac{\cancel{\sin x}}{\cos x} \cdot \frac{1}{\cancel{\sin x}}}$$

$$= \frac{\cos^3 x}{-2} = -\frac{1}{2} \checkmark$$



57) Evaluate $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 2}}{\sqrt{2x^2 + 1}}$.

estimate
 $\approx \frac{\sqrt{x^2}}{\sqrt{2x^2}} \approx \frac{1}{\sqrt{2}} \checkmark$

~~$$\begin{array}{l} \text{1st} \lim_{x \rightarrow \infty} \frac{2x}{2\sqrt{x^2 + 2}} \\ \hline \frac{4x}{2\sqrt{2x^2 + 1}} \end{array}$$~~

let $(y)^2 = \left(\frac{\sqrt{x^2 + 2}}{\sqrt{2x^2 + 1}} \right)^2$

$$y^2 = \lim_{x \rightarrow \infty} \frac{x^2 + 2}{2x^2 + 1} = \frac{\infty}{\infty}$$

$y^2 \stackrel{\text{1st}}{=} \lim_{x \rightarrow \infty} \frac{2x}{4x} = \frac{1}{2}$

$$y^2 = \frac{1}{2}$$

$$\sqrt{y^2} = \pm \sqrt{\frac{1}{2}}$$

$$y = \pm \frac{\sqrt{1}}{\sqrt{2}}$$

$$y = \frac{1}{\sqrt{2}} \checkmark$$

y cannot be negative
 because the equation
 is made of square roots.

58) Evaluate $\lim_{x \rightarrow -\infty} x^2 e^x$.

$\infty \cdot 0$

$$\lim_{x \rightarrow -\infty} \frac{x^2}{e^{-x}} \quad \frac{\infty}{\infty}$$

$$\stackrel{L'H}{=} \lim_{x \rightarrow -\infty} \frac{2x}{e^{-x}(-1)}$$

$$\stackrel{L'H}{=} \lim_{x \rightarrow -\infty} \frac{2}{e^{-x}}$$

$$= 0 \quad \checkmark$$

